

Literature Review:

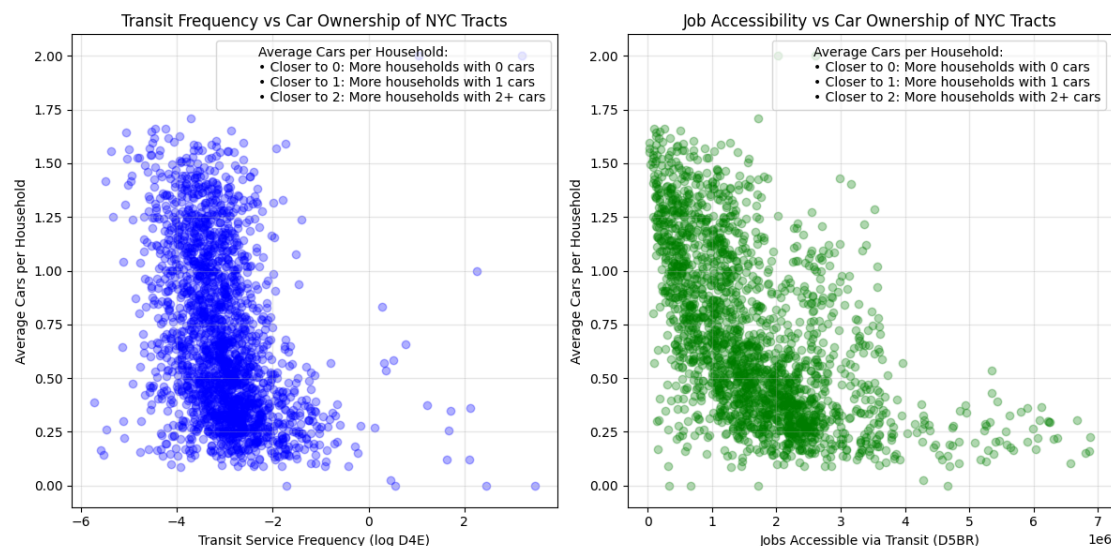
The relationship between transit service quality, job accessibility, and car ownership remains a critical area of inquiry in urban transportation policy. Studies have shown that the US experiences a relatively strong car dependency such that few car owners use transit, and that lack of access to the household vehicle(s) typically explains the choice of transit. As such, public transit's competition with cars heavily relies on service attributes such as frequency, reliability, and speed, as illustrated through the study of Los Angeles [1]. For example, when all else is equal, reducing in transit-to-auto travel time ratio increases the odds of selecting transit by about 25% and lowering the standard deviation of schedule deviation from over to under three minutes can result in 2.6 times increase in the odds [1]. The study finds that fast, frequent and reliable transit service along with fewer transfer requirements strongly correlate with car-owners' transit mode choice. This section aims to analyze whether sufficiently frequent and useful (for accessibility to jobs) transit services are capable of promoting transit services (to an extent that reduces car ownership) in NYC census tracts.

Methodology

The process began by preparing and filtering data from the Smart Location Database, specifically focusing on New York State (using `STATEFP='36'`) and narrowing to New York City's five boroughs (county FIPS codes: 005, 047, 061, 081, 085). Key variables were selected to represent transit infrastructure and car ownership: `D4E` (transit frequency per capita), `D5BR` (jobs accessible via a 45-minute transit commute), and household-level car ownership metrics (`AutoOwn0`, `AutoOwn1`, `AutoOwn2p`).

The data underwent cleaning to remove invalid entries, such as rows with negative values for transit frequency, job accessibility, or car ownership variables, to ensure only valid data points are analyzed. The dataset was then aggregated by tract and county, calculating mean values for each variable. Then, the "Total_Households" (summing all car-owning households) and "Avg_Cars" (a tract-level average of cars per household) variables were calculated. After compiling all the necessary variables, the following scatter plots were constructed.

Data Product: Does higher transit service frequency and job accessibility via transit correlate with reduced car ownership in NYC Tracts?



In the “Transit Frequency vs. Car Ownership” plot, the logarithmic transformation of `D4E` (transit service frequency) on the x-axis reveals a very vague negative trend with `Avg\_Cars` (average cars per household) on the y-axis. Most of the points are concentrated between the range of -4 and -2 on the x-axis while a few areas with higher transit frequency exhibit lower average car ownership, particularly in tracts where $\log(D4E)$ exceeds -2.

The “Job Accessibility vs. Car Ownership” plot, which maps `D5BR` (jobs accessible via transit) against “Avg_Cars” owned per household, shows a relatively more clear downward trend. Tracts with greater numbers of jobs accessible through transit tend to cluster toward lower average car ownership, particularly where `D5BR` exceeds 500,000 jobs. However, the relationship is nonlinear: at very high levels of job accessibility (e.g., over 1 million jobs), car ownership plateaus rather than continuing to decline.

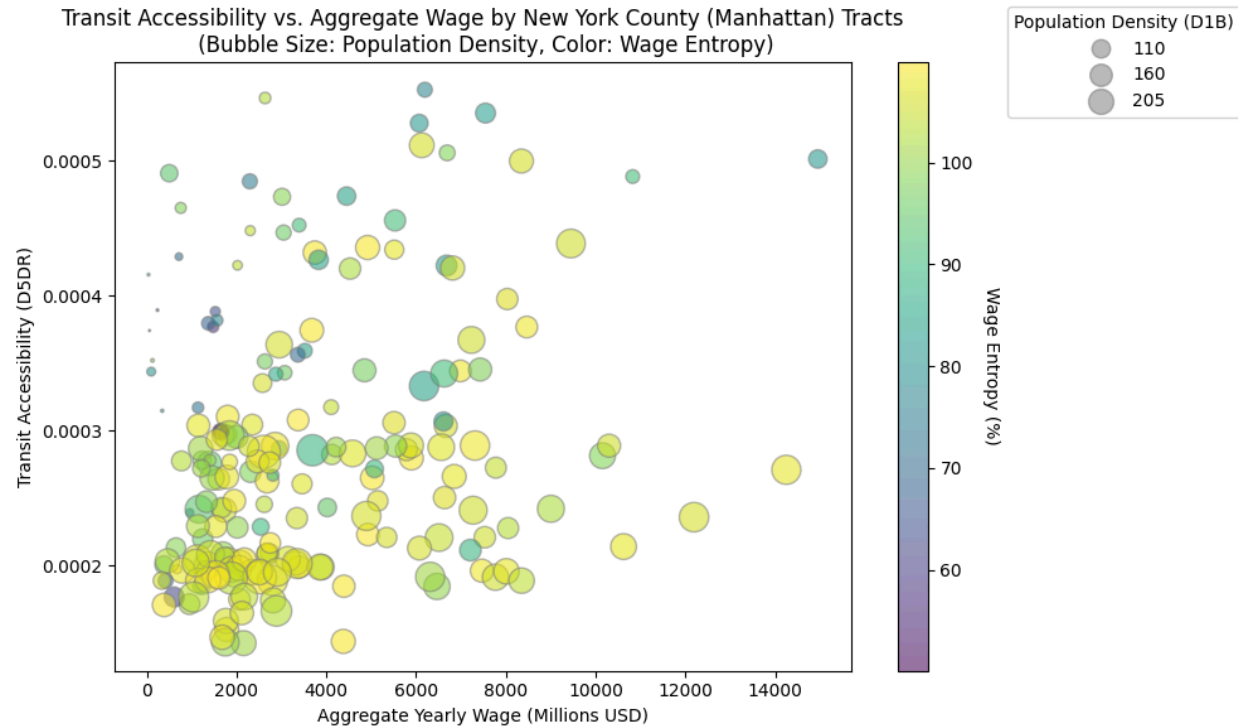
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The relationship between income levels, neighborhood income diversity, and job accessibility via transit is complex. Many studies have shown that car ownership declines sharply with income: in Washington, D.C., only 40% of low-income workers (<30% AMI) have access to a car, compared to over 80% of high-income workers (>120% AMI) [2]. This disparity suggests that low-income neighborhoods are more reliant on public transit. However, according to the article, geographic distribution of transit infrastructure exacerbates inequities between low-income and high-income neighborhoods [2]. Urban areas offer more robust public transit systems, which correlate with lower car ownership, while rural low-income counties have limited transit options and high poverty rates, forcing residents to depend on cars, which they often cannot afford [2]. This implies that residents in low-income neighborhoods—particularly rural ones—have fewer jobs accessible via transit compared to high-income urban areas with established transit networks and, consequently, greater job accessibility. This section aims to extend this research to census tracts in New York county (Manhattan) to assess whether the trend identified holds true in New York county.

Methodology

The process begins by accessing and cleaning the necessary variables: wage metrics (“W_P_Lowwage”, “W_P_Medwage”, “W_P_Highwage” and “Aggregate_Wage”) and transit-based job accessibility (“D50R”). The dataset is filtered to focus on New York counties (e.g., Manhattan, coded as `061`), with constraints applied to ensure valid wage percentages (0–1) and non-negative accessibility values, to eliminate nonsensical data, such as negative accessibility scores. Using the wage metrics, the “income entropy” is calculated such that higher entropy indicates greater income diversity. Finally, after combining all the necessary data, a bubble plot is generated to visualizes relationships: aggregate wages (x-axis) and transit accessibility (`D50R`, y-axis) are plotted, with bubble size reflecting population density (`DIB`) and color encoding entropy.

Data Product: Do residents in high income or income-entropy neighborhoods have greater access to jobs via transit (greater number of regional jobs accessible by transit) than those in low income or low-entropy neighborhoods in New York County (Manhattan)?



The bubble plot illustrates multidimensional relationships between economic factors, transit accessibility, and population density across New York County (Manhattan) tracts. There appears to be a very vague positive relationship between aggregate yearly wage (economic output) and transit accessibility (D5DR) such that tracts with higher aggregate wages generally correspond to higher transit accessibility for relatively large (greater population density wise) tracts.

Citations

[1] Chakrabarti, Sandip. "How can public transit get people out of their cars? An analysis of transit mode choice for commute trips in Los Angeles." *Transport Policy*, vol. 54, Feb. 2017, pp. 80–89, <https://doi.org/10.1016/j.tranpol.2016.11.005>.

[2] Mengedoth, Joseph. "Transportation Access as a Barrier to Work." *Federal Reserve Bank of Richmond*, 2023, www.richmondfed.org/publications/research/econ_focus/2023/q4_district_digest.